

Ashok Sasitharan

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TECHNICAL SKILLS

Languages: Ruby | Python | Postgres | MySQL | JavaScript | HTML/CSS

Frameworks: Ruby on Rails | ReactJS | Node.js | Flask | Next.JS

Developer Tools: Git | Docker | Nginx

EXPERIENCE

Software Engineer

Apr 2022 – Nov 2025

Bell Canada

Mississauga, ON

- **Owned end-to-end development** of a **device certification system** integrating an **Anritsu network simulator**, enabling **automated signal testing** required for Android devices prior to market release
- **Maintained and extended** a **Ruby on Rails**-based device automation platform used by multiple internal teams, contributing new features, improving reliability, and supporting ongoing production use
- **Authored over 60 automated test scripts** for use in our automation platform that performs Android and web automation tasks to ensure timely certification of upcoming devices thereby reducing manual certification effort by **2,000 hours** and saving an estimated **\$200,000 annually**
- **Designed and built** a dockerized **AI microservice** (Flask) leveraging image detection and Tesseract OCR to identify UI elements during automation runs; made all backend architecture and API design decisions
- **Exposed REST APIs** to integrate the AI microservice with the automation framework, returning actionable coordinates for programmatic interaction and improving test robustness
- **Provided production support** for a mission-critical internal testing platform, debugging failures, assisting partner teams, and minimizing delays to device release timelines

PROJECTS

Badminton Stat Tracker | *React, Express, SQLite*

- **Designed and built** a mobile-first **PWA** for creating groups, scheduling sessions, and tracking player stats with **live singles and doubles leaderboard**
- **Developed** a **dynamic matchmaking engine** for singles/doubles, managing uneven players, sit-outs, rotations, and **accurate stat tracking**
- Designed **relational data models** and REST APIs with role-based access and transactional enrollment, eliminating **race conditions** and **preventing overbooking**
- **Managed deployment** and operations, self-hosting via **Docker** and **CI/CD pipelines** to **automate testing and production releases**

Travel Planner | *React, Typescript, Rails, PostgreSQL*

- **Architected** a normalized trip-planning domain model (Trips → Days → Items) and implemented transactional APIs to ensure data consistency during reordering, reassignment, and deletion operations
- Engineered a **drag-and-drop scheduling system** with optimistic UI state management and order normalization to maintain deterministic client/server state under frequent mutations
- Designed the system for extensibility, enabling future features such as **calendar-based rendering**, **expense splitting**, and **collaborative editing** without restructuring core domain logic

EDUCATION

Durham College

Ontario College Advanced Diploma, Computer Programming and Analysis

Oshawa, ON

Sept 2019 – Apr 2022